

Lab Assignment: Skin Lesion Boundary Detection Using Canny Edge Detection

Problem Statement

Develop a simple computer vision system that detects the **boundary of a skin lesion** from a skin image using image filtering and Canny edge detection.

The goal is to determine how effectively edge detection can separate the lesion from the surrounding skin.

Lab Tasks

Task 1 — Load the Image

- Select a skin lesion image from a public dataset or instructor-provided images.
- Display the original image.

Task 2 — Preprocess the Image

- Convert the image to grayscale.
- Apply a **Gaussian filter** to reduce noise.
- Display the grayscale and filtered images.

Task 3 — Apply Canny Edge Detection

Apply Canny edge detection using **three different threshold settings**, for example:

- 50–100
- 100–200
- 150–250

Display all three edge maps.

Task 4 — Select the Best Result

Compare the three edge maps and select the threshold that provides the **clearest lesion boundary**.

Explain why you selected it.

Task 5 — Detect the Lesion Boundary

Use the detected edges to obtain the lesion boundary using a suitable technique such as:

- Contour detection
- Morphological operations
- Thresholding

Draw the detected boundary on the original image.

Task 6 — Calculate Lesion Area

Calculate the approximate number of pixels inside the detected lesion boundary.

Report:

- Lesion area
- Lesion perimeter

Required Visualization

Your final output must contain:

Original → Grayscale → Gaussian Filter → Canny → Lesion Boundary

Calculate:

- **Lesion area**
- **Lesion perimeter**

Present the results in the following table:

Image	Best Filter	Edge Method	Area (pixels)	Perimeter (pixels)
Image 1				
Image 2				
Image 3				
Image 4				
Image 5				

Final Comparison

Must complete the following table.

Method	Noise Handling	Edge Quality	Boundary Detection	Overall Performance
Original + Sobel				
Original + Canny				
Average + Sobel				
Average + Canny				
Gaussian + Sobel				
Gaussian + Canny				
Median + Sobel				
Median + Canny				

Questions to Answer

1. Why is Gaussian filtering applied before Canny detection?
2. How did the three Canny threshold settings affect the result?
3. Which threshold produced the best lesion boundary?
4. Why are edges useful for detecting skin lesions?
5. What problems did you observe in detecting the lesion boundary?
6. How could your method be improved?